Pegasus II ISS operating manual

Step 1. Tripod and Antenna's.
It is assumed the tripod is well anchored and the mast is reasonably vertically "straight".
See "Pegasus II antenna setup manual."
1.1] Mast stay cables that secure the antennas from wind. (Not the tripod anchors!)
Loosen the mast anchor stays one by one; at the distant anchor points. Secure each, in sequence of loosening, to the lower mast stay locking ring. Make sure that the mast stay cables do not cross any wires or coaxes. The mast stay cables must not cross each other. Pull the locking ring down to tighten the mast stay cables.
1.2] Wire and coax freedom to move more than 360 degrees.
Loop the coax, rotator control cables, direction sense cables, and antenna polarity control wires.
In large circles, directly below the VHF and UHF yagi antennas.
Be careful that the wires do not twist around the mast, too much.
1.3] Check all coax and cable connections and terminations.
1.4] Apply AC power to the remote "toolbox" control unit.

Step 2. Pegasus II controller.
Leave out the quotes, below. [space] means 1 space.
2.1] Power connections.
12 Volts DC to the controller. 5 Volts DC to the LED matrix
2.2] USB connection to the PC (etc) with a serial monitor program.
Allows the monitoring and control of Pegasus.
From the PC's serial monitor: Baud rate 19200. Line ending CRLF.
Use the command PS [space] "2.4G WIFI SSID" to join a WIFI network. Persisted.
Use command PP [space] "password" to set the WIFI password. Persisted.
Use command RST to reset the controller.
2.3] WIFI connections.
2.3.1] The Pegasus II **dashboard** is accessible from any browser on any device in the same WIFI network.
The **dashboard** shows controller responses and allows controller commands to be sent.
To access the Pegasus II **dashboard**, use address zs6peg.local. ("zs6peg" is the host name.)
The userid is admin. The password is admin.
A list of all possible commands can be shown. Click on 'HELP' (button turns green). Click on 'CMD'.
2.3.2] The Pegasus II **monitor** is accessible from any browser on any device in the same WIFI network.
The **monitor** shows 10 second updated status when tracking a satellite. The radio PTT Transmit / Receive status is shown and can be remotely controlled.
The Pegasus II **monitor** address is zs6peg.local/monitor.

Step 3. Satellite tracking preparation.
Use the PC serial monitor and/or the Pegasus II WIFI dashboard.
3.1] Current position and Satellite identity commands.
AS1 display longitude, latitude and altitude. Persisted after power off.
AS2 set longitude e.g. "AS2 28.13511"
AS3 set latitude e.g. "AS3 -26.2719"
AS4 set altitude in meters e.g. "AS4 1500"
AS5 display catalogue number. Persisted after power off.
AS6 set ISS default catalogue number 25544
AS7 set other catalogue number e.g. "AS7 40967"
3.2] Two Line Element Data update commands.
AS8 get TLE data from internet. Celestrak. Persisted after power off.
AS9 show TLE elements age.

3.3] Set Antenna NORTH direction commands.
S - Stop All Rotation. Or from the WIFI dashboard the STOP button.
L - Rotate Azimuth CCW
R - Rotate Azimuth CW.

H - Ham4 Super North Offset info.
H3 - Reset North override to OFF.
H2 - Set North here.

C3 - get ADC degrees.

3.4] Radio CAT control commands. (ICOM IC-9700 only.)
D4 - Get/Set center TX radio frequency in Hertz. e.g. D4 145990000. Persisted after power off.
D5 - Get/Set center RX radio frequency in Hertz. e.g. D5 437800000. Persisted after power off.

ASg Set / Get AOS to LOS total Doppler frequency VHF range. e.g. ASg or ASg 6000.
Persisted after power off.
ASi Set / Get AOS to LOS total Doppler frequency UHF range. e.g. ASi or ASi 17000.
Persisted after power off.

3.5] Antenna switches control commands.
Y1 is TX on VHF (Relay normally off in RX). For the ISS FM repeater.
Y3 is TX and RX on VHF (Relay always on). For DIRECT ISS contacts.
Note: the antennas can also be connected permanently to the VHF yagi.
Y4 is TX and RX on UHF (Relay always off). For ISS SSTV.
Note: the antennas can also be connected permanently to the UHF yagi.

3.6] ISS Satellite PASS prediction for latest collected TLE data.
Provides enough data for ISS PASS selection.
APxxx Predict next xxx passes. E.G. AP009.
A File called "Predict.txt" is created on the SD card with this data.

3.7] Specific ISS PASS detail. Also used to define settings that change for each pass.
NOTE: Even though the satellite tracking starts by re-doing this step.
NOTE: It is important for preparation.
NOTE: This information is NOT Persisted. Needs to be re-done after power off.
AQ predict one pass after date/time. e.g. AQ 2026 07 20 21 44 30
A file called "PredictOne.txt" is created on the SD card with this data.

Example:
Epoch: 29/8/2026 10:52:60.00

Month = 8 Day = 29
Start: az=318.85° 10:55:16.47
Max: elev=88.05° 11:0:43.52
Stop: az=138.29° 11:6:16.19
Stop: az=138.29° 11:6:16.19 Daylight

AOS to LOS = 660 seconds, or
11 minutes and 0 seconds
10 sec doppler delta = 90 Hertz.
10 sec UHF doppler delta = 120 Hertz.
Sat turn direction = CCW
Pre Start degrees > 318.85
Time: 32 milliseconds
Prediction complete.

3.8] Antenna Pre-position.
According to 3.7] turn the antennas to about 4 to 8 degrees past or before the
"Pre Start Degrees". Note this setting changes for each satellite pass.
Use the "L", "R" and "S" commands for coarse adjustment.
USE command "S2" to enable the rotators automatic measurement.
Command "S0" shows the status of this and some other switches.
Fine adjustment - Less than 10 degrees.
Command Mxxx will automatically move the Azimuth rotator until "xxx" degrees.
Command E2xxx will automatically move the Elevation rotator until "xxx" degrees.

3.9] Radio Preparation. (ICOM IC-9700 only.)
Check that the serial CAT cable is connected to the radio.
3.9.1] Turn ON the enable radio cat control switch. Command "S9".
3.9.2] Load the radio with command "D6".
3.9.3] For emergency operation, Load the ONE minute doppler Memories. Command "DC".

Step 4. Live Satellite tracking.
About 10 or 15 minutes prior to the satellite pass, use command "ASb" to set time to current time and start tracking.
Use the WIFI monitor program to see the pass's status updates.

Use command "ASc" to force the satellite tracking routine to stop.

Use command "AY2" to Toggle satFileLogging. Tracking data will be recorded to SD file "satellite_log.txt".

Replay past or future ISS passes, with tracking.
Load the "tle.txt" file that was used, or will be used, onto the SD card. If the TLE data has not changed since the last update, this is not required. Instead of Step4. do the following.
Note the start time of the ISS pass to be tracked. Use the following command with a time stamp about 2 or 5 minutes before the pass start time.
"ASa". Set future time manually and start tracking. e.g. "ASa 2025 10 05 05 55 00".
Year Month Day Hour Minute Second.

Use command "ASc" to force the satellite tracking routine to stop.

Use command "AY2" to Toggle satFileLogging. Tracking data will be recorded to SD file "satellite_log.txt".